Q.28 Explain the implementation of AND gate using a ladder diagram.

Q.29 What is the function of SFRs in 8051 microcontroller?

Q.30 Explain any two comparison instructions for PLCs.

Q.31 Write a note on memory structure of PLCs.

Q.32 What is ladder diagram? Explain with any one example.

Q.33 How many input/output ports are there in 8051 microcontroller? Explain the structure on any one port.

Q.34 Write a short note on addressing modes on 8051 microcontroller.

Q.35 What is interfacing? Describe any one example of interfacing with 8051 microcontroller.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Draw and explain the pin diagram of 8051 microcontroller.

Q.37 Explain the architecture of PLC in detail using a block diagram.

Q.38 Write notes on any two

a) Timer instructions in PLC

b) Interfacing LCD with 8051 microcontroller

c) Assembler directives in 8051 microcontroller.

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Roll No. ....

6th Sem / Elect, Eltx, Mecatronics, GE, Power Station Engg, Elect, & Eltx Engg.

Subject:- Programmable Logic Controllers and Microcontrollers / Mic. Cont. & PLCs

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 An OR function implemented in ladder logic uses:

a) Normally-closed contact in series

b) Normally-open contacts in series

c) A single normally-closed contact

d) Normally-open contacts in parallel

Q.2 Interrupts which can be ignored are known as

a) Maskable Interupts

b) Non-maskable interrupts

c) Vectored Interrupts

d) Non-vectored Interrupts

Q.3 PLC operates on the following signals

a) Only analog signals

b) Only digital signals

c) Both analog and digital signals

d) Only serial communication signals

Q.4 What does PLC stand for?

a) Programmable Logic Computer

b) Programmable Language Controller
c) Programmable Logic Controller
d) Process Logic Control

Q.5 The programmable logic controllers have applications in________
a) Paint industry b) Packaging industry
c) Bottling plants d) All of the above

Q.6 Which port of the 8051 is known as a "true bidirectional port" and requires external pull-up resistors?
a) Port 1 b) Port 2
c) Port 0 d) Port 3

Q.7 Which of the following is an example of a discrete output device controlled by a PLC?
a) Variable speed drive b) Proportional valve
c) Solenoid valve d) Analog meter

Q.8 How many data pins does the 8051 microcontroller have?
a) 8 b) 16
c) 32 d) 40

Q.9 What is the size of the on-chip ROM (Program Memory) in the standard 8051?
a) 64 KB b) 128 KB
c) 4 KB d) 256 bytes

Q.10 A Counter Up (CTU) instruction in PLC increments its accumulated count each time its input goes from:
a) TRUE to FALSE b) FALSE to TRUE
c) Remains FALSE d) Remains FALSE

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Write any two applications of microntrollers.

Q.12 CPU stands for________.

Q.13 8051 micontroller has______number of input / Output ports)

Q.14 FBD stands for_________.

Q.15 ________ module in a PLC sends control signals to devices (like motors or lights).

Q.16 Which company manufactures PIC microntrollers?

Q.17 The ________in a PLC is used to provide a safe and reliable power source for the system.

Q.18 In 8051 microcontroller SFR stands for_______.

Q.19 In 8051 microconntrollers PC stands for________.

Q.20 ROM stands for_________.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write any five applications of microcontrollers.

Q.22 Name any 5 manufacturers of PLCs.

Q.23 Explain the use of PLC in home automation.

Q.24 Write a short note on sequencers.

Q.25 Explain RTC instruction in PLCs.

Q.26 Explain why PLCs have replaced relays in automation?

Q.27 Write a short note on 8051 interrupts.